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Focus measuring arrangement and method of ophthalmic examination, and ophthalmic examination apparatus and method

Abstract

A focus measuring arrangement for ophthalmic examination comprises an infrared light source system, which directs infrared light to a pupil of an eye of a person and forms a focusing pattern of the infrared light on a retina. A receiving arrangement apertures receives the infrared light of the focusing pattern from the retina through separate areas of the pupil. An image forming arrangement causes the infrared light passed through the apertures to form an individual image of the focusing pattern for each of the apertures in an image space of the receiving arrangement, the image forming arrangement causing an optical correspondence between the at least two apertures and the separate areas of the pupil. A detecting surface arrangement is located at a non-zero distance behind the at least two apertures and the image forming arrangement in a direction of propagation of the infrared light reflected from the retina for receiving the infrared light passed through the at least two apertures. A data processing unit forms focusing adjustment information based on locations of illuminated spots of the infrared light of the focusing pattern on the detecting surface arrangement, and outputs the focusing adjustment information for further processing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national phase of International Application No. PCT/FI2021/050874 filed Dec. 14, 2021 which designated the U.S. and claims priority to FI patent application No. 20206351 filed Dec. 21, 2020, the entire contents of each of which are hereby incorporated by reference.

FIELD

(2) The invention relates to a focus measuring arrangement for ophthalmic examination, an ophthalmic examination apparatus, a focus measuring method of ophthalmic examination and an ophthalmic examination method.

BACKGROUND

(3) A typical problem with ophthalmic instruments is how to focus the ophthalmic instrument to patient's eye. The problem is particularly difficult with examination instruments which are used to examine optically portions of the eye behind the iris, such as a fundus cameras. A wrong focus leads to a bad image quality of the retina, for example. Challenges in focusing are caused by a short time period available for focusing, and changes in eye focus i.e. continuous accommodation, for example. Further challenges may be caused by eye disorders such as myopia, hyperopia or astigmatism, or by eye diseases and by eyes operated by laser eye surgery. Hence, an improvement would be welcome.

(4) Patent document US20140168401 presents a system and a method for remote measurement of optical focus. Patent document US20120154748 presents a fundus camera. Patent document US20110007272 presents also a fundus camera. Patent document US20150070655 presents an apparatus and a method for automatic alignment in an optical system and applications. Patent document US20120002167 presents an ophthalmic apparatus. Patent document US20200205659 presents an ophthalmic apparatus and a method for controlling the same. Patent document EP2243421 presents an ophthalmic device.

BRIEF DESCRIPTION

(5) The present invention seeks to provide an improvement in the focusing of the ophthalmic examination instrument to eye.

(6) The invention is defined by the independent claims. Embodiments are defined in the dependent claims.

Description

LIST OF DRAWINGS

(1) Example embodiments of the present invention are described below, by way of example only, with reference to the accompanying drawings, in which

(2) FIGS. 1A and 1B illustrate examples of a focus measuring arrangement and an ophthalmic examination apparatus for ophthalmic examination;

(3) FIG. 2A illustrates an example of three illuminated spots on a single detecting surface

arrangement;

(4) FIG. 2B illustrates an example of two illuminated spots on separate detecting surfaces of the detecting surface arrangement;

(5) FIG. 3 illustrates an example of the focus measuring arrangement where focusing spots are directed to an eye with a beam splitter;

(6) FIG. 4 illustrates another example the focus measuring arrangement where focusing spots are directed to an eye with a mirror;

(7) FIG. 5 illustrates an example of the ophthalmic examination apparatus;

(8) FIGS. 6A and 6B illustrate examples of paths of infrared light of a focusing pattern, the infrared light going through separate sections at an aperture stop;

(9) FIG. 7A to 7D illustrate examples of the focusing patterns;

(10) FIG. 8 illustrates examples of rays of infrared light and their locations of illuminated spots on the detecting surface arrangement as a function of a state of focusing (in-focus or out-of-focus);

(11) FIG. 9A to 9D illustrate examples of various focusing patterns and their paths at the pupil of the eye and/or the illuminated spots at the detecting surface arrangement;

(12) FIGS. 9E to 9G illustrate examples of illuminated spots of various focusing patterns at a detecting surface arrangement and detecting surfaces;

(13) FIG. 10 illustrates an example of a data processing unit;

(14) FIG. 11 illustrates an example of a flow chart of a measuring method of ophthalmic examination; and

(15) FIG. 12 illustrates of an example of a flow chart of an ophthalmic examination method.

DESCRIPTION OF EMBODIMENTS

(16) The following embodiments are only examples. Although the specification may refer to “an” embodiment in several locations, this does not necessarily mean that each such reference is to the same embodiment(s), or that the feature only applies to a single embodiment. Single features of different embodiments may also be combined to provide other embodiments. Furthermore, words “comprising” and “including” should be understood as not limiting the described embodiments to consist of only those features that have been mentioned and such embodiments may also contain features/structures that have not been specifically mentioned. All combinations of the embodiments are considered possible if their combination does not lead to structural or logical contradiction.

(17) It should be noted that while Figures illustrate various embodiments, they are simplified diagrams that only show some structures and/or functional entities. The connections shown in the Figures may refer to logical or physical connections. It is apparent to a person skilled in the art that the described apparatus may also comprise other functions and structures than those described in Figures and text. It should be appreciated that details of some functions, structures, and the signalling used for measurement and/or controlling are irrelevant to the actual invention. Therefore, they need not be discussed in more detail here.

(18) First, here are some basic information and working principles of an ophthalmic instrument. The optical instrument comprises at least one image plane, and an imaging optics, where during the use of the ophthalmic instrument, at least a portion of the retina needs to be imaged to the at least one image plane using the imaging optics. Or, the same can be expressed in another way: each of the at least one image plane needs to be imaged on the retina. Here, an image plane should be adjusted be at a detector cell of a digital camera.

(19) The image plane may be in virtual space, i.e. the image formed may be virtual instead of real image. The image plane may not be a physical object, but may be a predetermined surface in space. The image plane may locate outside the ophthalmic instrument, and may be far away or in infinity. The image plane may not be necessarily a plane, but may be a curved surface.

(20) In order to achieve the imaging condition in a sufficient accuracy, the imaging optics needs to be focused to the retina. For example, when using the fundus camera, the fundus camera optics needs to be focused to the retina in order to capture a sharp image of the retina on the camera

sensor.

(21) The ophthalmic instrument, which comprises a focus measuring arrangement described in this document, refers to any optical instrument (for example used for examination or treatment of eye) which requires focusing to the retina.

(22) The ophthalmic instrument contains an imaging optics (lenses/mirrors etc.), and the at least one imaging plane which is or is imaged on a camera sensor of a fundus camera. During the use of the ophthalmic instrument, there is a need to focus the imaging optics so that it images at least portion of the retina to the imaging plane using for example ultraviolet (UV), visible (VIS), or near infrared (NIR) wavelengths. The pupil stop of the imaging optics may be imaged, by at least portion of the imaging optics, to a real image, called a first pupil, into between the retina and the ophthalmic instrument. Let us define pupil plane **152'** as plane perpendicular to the optical axis of the imaging optics and going through the first pupil.

(23) FIG. 1A illustrates an example of a focus measuring arrangement for ophthalmic examination. In the example of FIG. 1A, the focus measuring arrangement is out-of-focus. In this kind of situation focused individual images **124A**, **124B** of the focusing pattern **130** that are formed through at least two apertures **120A**, **120B** are in the imaging space **110** of the receiving arrangement **102** but not on a detecting surface arrangement **106**, and that is why in this example, the focused individual images **124A**, **124B** are in front of the detecting surface arrangement **106**. However, focuses of the images **124A**, **124B** may as well be behind the detecting surface arrangement **106** in an out-of-focus situation. The ophthalmic instrument comprising the focus measuring arrangement is in the examination position or the examining range for examining the eye **150**. The focusing pattern **130** may comprise at least one line, at least one point, at least one array of lines or the like without limiting to these. The focusing pattern **130** may increase contrast variations compared to bare retina **154**.

(24) The at least one of the at least two apertures **120A**, **120B** may be located in the image space **110** at a non-zero distance from an optical axis OA of the receiving arrangement **102**.

(25) The focus measuring arrangement comprises an infrared light source system **100**, which directs infrared light to a pupil **152** of an eye **150** of a person **156** when it is within an examining range from the eye **150**. Note that the pupil plane **152'** is at the pupil **152** or at least approximately at the pupil **152** when the ophthalmic instrument is focused and ready for the examination of the eye **150**. The infrared illumination may be mostly or fully invisible to the eye **150**, so that it does not disturb the eye **150** during the measurement of focus and during the examination. For example, the most of the power of the infrared illumination may be between 700 nm and 1000 nm spectral wavelength range. The infrared illumination may also contain some power in the visible spectral region, i.e. between 400 and 700 nm wavelength.

(26) In an embodiment, the infrared light source system **100** may comprise a light source **10**, which outputs the infrared light with the focusing pattern **130** (see FIGS. 1A, 1B, 3 and 4).

(27) The infrared light source **100** also forms the focusing pattern **130** of the infrared light on a retina **154** of the eye **150** (the focusing patterns **130** are illustrated in FIGS. 7A to 7D). In an embodiment, the infrared light source system **100** may comprise a first beam splitter **200**, which turns at least a portion of the infrared light toward a desired direction. The desired direction is toward the eye **150** when the focus measuring arrangement is within the examining range from the eye **150**. The first beam splitter **200** allows at least a portion of the infrared light reflected from the retina **154** pass through toward a detecting surface arrangement **106** in an image space **110** of the receiving arrangement **102**. The receiving arrangement **102** comprises optical components that gather light coming from the retina **154** through the pupil **152** of the eye **150**, and focus the light on the detecting surface arrangement **106**. A part of the optical components may be common to the imaging system **300** of the ophthalmic examination apparatus.

(28) The first beam splitter **200** can be considered common to the infrared light source system **100** or the image forming arrangement **104**. The first beam splitter **200** may receive the infrared light

and reflect the infrared light at least partly to the pupil **152** of the eye **150**, and allow the infrared light reflected from the retina **154** to propagate through at least partly toward the receiving arrangement **102**.

(29) The image forming arrangement **104** may comprise one of more optically refractive and/or reflective components which may form a real image or a virtual image. The one of more optically refractive and/or reflective components may comprise at least one lens and/or mirror, which has a curved surface.

(30) The focus measuring arrangement comprises a receiving arrangement **102**, which in turn comprises at least two apertures **120A**, **120B**, which receive the infrared light of the focusing pattern **130** from the retina **154** through separate areas **122A**, **122B** of the pupil **152**. This makes the receiving arrangement **102** to operate in a non-telecentric manner. They may be at the same or a different distance from the detecting surface arrangement **106**.

(31) The focus measuring arrangement also comprises an image forming arrangement **104** in the receiving arrangement **102**, at least some part of which causes the infrared light passed through the at least two apertures **120A**, **120B** to form the individual images **124A**, **124B** of the focusing pattern **130** for each of the at least two apertures **120A**, **120B** in the image space **110** of the receiving arrangement **102**. Only a single the individual image **124A**, **124B** is formed through one of the apertures **120A**, **120B**, and there is only one individual image **124A**, **124B** per a single aperture **120A**, **120A**. The image forming arrangement **104** additionally causes an optical correspondence between the at least two apertures **120A**, **120B** and the separate areas **122A**, **122B** of the pupil **152**. That is, at least a part of the image forming arrangement **104** causes the optical correspondence. The optical correspondence may mean that the image forming arrangement **104** forms images of the at least two apertures **120A**, **120B** on, at or directly adjacent to the pupil **152** of the eye **150**, the images being the areas **122A**, **122B**. The images may be formed, in practice and/or in theory, which makes the areas **122A**, **122B** and the apertures **120A**, **120B** have the optical correspondence. The image forming arrangement **104** comprises at least one optical component which can form a real image, the optical component being a lens or a mirror, for example.

(32) The focus measuring arrangement further comprises the detecting surface arrangement **106** located at a non-zero distance behind the at least two apertures **120A**, **120B** and the image forming arrangement **104** in a direction of propagation of the infrared light reflected from the retina **154**.

(33) The detecting surface arrangement **106** has an image plane thereon when the focus measuring arrangement is in-focus. The image of at least portion of the retina **154** is formed on it. The detecting surface arrangement **106** may comprise a matrix sensor such as a complementary metal oxide semiconductor (CMOS) or charge coupled device (CCD) sensor. The detecting surface arrangement **106** may also be a reticle, a slit or an input head of an optical fiber or cable.

(34) In an embodiment, the detecting surface arrangement **106** may comprise only one detecting surface **128** (see FIG. 2A), which may be common to the infrared light from the at least two apertures **120A**, **120B** for receiving the infrared light passed through the at least two apertures **120A**, **120B**.

(35) In an embodiment, the detecting surface arrangement **106** may comprise a number of detecting surfaces such that the number of detecting surfaces is smaller than a number of at least two apertures **120A**, **120B**, and each of the detecting surfaces receives the infrared light from at least one of the at least two apertures **120A**, **120B** (not illustrated in Figures).

(36) In an embodiment, the detecting surface arrangement **106** may comprise at least two detecting surfaces **128**, **129**, each of which receives the infrared light from a single aperture of the at least two apertures **120A**, **120B**. In that manner, there is one to one correspondence between the apertures **120A**, **120B** and the detecting surfaces **128**, **129**.

(37) The focus measuring arrangement finally comprises a data processing unit **108**, which forms focusing adjustment information based on locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of illuminated spots **126A**, **126B** of the infrared light of the focusing pattern

130 on the detecting surface arrangement **106**. The data processing unit **108** then outputs the focusing adjustment information for further processing.

(38) In an embodiment, the data processing unit **108** may determine locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of centers of the illuminated spots **126A**, **126B**, **126C**. In an embodiment, the data processing unit **108** may determine locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of centers based on image processing.

(39) FIG. 1A also shows an examination detector cell **302** of an imaging system **300** of an ophthalmic examination apparatus for which focusing adjustment information is computed. The ophthalmic instrument comprises both the focus measuring arrangement and the ophthalmic examination apparatus. The examination detector cell **302** has an optically determined relation with respect to the detecting surface arrangement **106**. They may be simultaneously in-focus, but alternatively they do not need to be simultaneously in-focus. It is the locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of the illuminated spots **126A**, **126B** of the focusing pattern **130** on the detecting surface arrangement **106** that have a relation to the focus of the examination detector cell **302** of an imaging system **300** of an ophthalmic examination apparatus.

(40) In the example of FIG. 1B, the focus measuring arrangement is in-focus or the spots are in locations (X.sub.0, Y.sub.0) which correspond to a state where the retina **154** is in-focus on the examination detector cell **302**. In this kind of situation properly focused individual images **124A**, **124B** of the focusing pattern **130** may be formed through at least two apertures **120A**, **120B** on the detecting surface arrangement **106**. In this example, the light, which may be visible light and reflected from the retina **154**, may also be in-focus at the examination detector cell **302**.

(41) The focusing may be performed with the focus measuring arrangement in the following manner.

(42) A) The ophthalmic instrument is brought in such position i.e. examination position with respect to the eye **150** that when illumination of the infrared light source **100** is on, the focusing pattern **130** reflected from the retina **154** is imaged, in-focus or out-of-focus, to the detecting surface arrangement **106**. This position is within the examination range of the ophthalmic instrument.

(43) B) The infrared light source **100** forms the focusing pattern **130** on the retina **154**. The focusing pattern **130** may contain illuminated features, i.e. contrast differences, on the retina **154**, or figures or contrast differences defined by structured illumination (see examples of FIGS. 7A to 7D). The images of the focusing pattern **130** are then captured by the detecting surface arrangement **106**. The images may at least initially be out-of-focus before a focus adjustment.

(44) C) Then a predetermined algorithm is used (which depends on the used arrangement) for calculating the required focus shift from the captured images of the focusing pattern **130**.

(45) D) The ophthalmic instrument is focused by the required amount in order to get the image of the retina **154** focused on the examination detector cell **302**.

(46) E) The focusing process of steps A) to D) or B) to D) may be repeated, if needed.

(47) FIG. 2A illustrates an example of three illuminated spots **126A**, **126B** and **126C** on the on the detecting surface arrangement **106**, which is in this example a single detecting surface **128**. The scale of the coordinate system is arbitrary in FIG. 2A. The spots **126A**, **126B** may be blurred or sharp. In an embodiment, the data processing unit **108** may form the focusing adjustment information based on a geometric characteristic of the at least two illuminated spots **126A**, **126B**, **126C** on the detecting surface arrangement **106**, the geometric characteristic being a function that relates to a deviation with respect to a state where the focus measuring arrangement is in-focus and/or the state where the retina **154** is in-focus on the examination detector cell **302**.

(48) The geometric characteristic refers to locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of the illuminated spots **126A**, **126B**, **126C** with respect to each other, while a certain locations (X.sub.A0, Y.sub.A0), (X.sub.B0, Y.sub.B0), (X.sub.C0, Y.sub.C0) of them may be used as a reference, on one or more detecting surfaces **128**, **129**. The locations (X.sub.A,

Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) and/or a shape or form formed by the locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of the illuminated spots **1206A**, **126B**, **126C** and its change can be determined mathematically in the data processing unit **108**.

(49) If a plurality of the more detecting surfaces **128**, **129** are used, they have fixed or known locations with respect to each other. The illuminated spots **126A**, **126B**, **126C** form a constellation (see line connecting the spots in FIG. 2A) that deterministically varies if a total refraction power of the eye **150** and the focus measuring arrangement varies. Typically the refraction of the eye **150** varies temporally and/or person by person **156** and the variation of the refraction can be monitored with the geometric characteristic of the illuminated spots **126A**, **126B**, **126C** on the detecting surface arrangement **106**. Only one configuration of the varying constellation the illuminated spots **126A**, **126B**, **126C** corresponds to a proper focus, where an image of the retina **154** is in-focus, for the ophthalmic examination apparatus. The geometric characteristic can be used to identify the state of focus in a deterministic and recognizable manner.

(50) In an embodiment an example of which is illustrated in FIG. 2B, the geometric characteristic may refer to at least one distance D between the at least two illuminated spots **126A**, **126B** on the detecting surface arrangement **106**. In this example the detecting surface arrangement **106** comprises two detecting surfaces **128**, **129**. The distance D may vary depending of the refraction thus revealing the need of the adjustment of focus. In this example, no focus has been reached and the dashed circles show the situation where focus has been reached. Hence, the dashed circles illustrate a reference characteristic linked to a proper focus.

(51) The location difference of the illuminated spots **126A**, **126B**, **126C** may be calculated by various methods. For example, a cross-correlation between the images of the focusing pattern **130** on the detecting surface arrangement **106** and the reference may be calculated, and determining the location shift between the images and the reference that gives a highest correlation. That can be calculated for example by determining locations where a derivative or a gradient of the cross-correlation is approximately zero or crosses zero.

(52) The location and/or geometric difference may also be calculated for example by pattern recognition.

(53) In general, the data processing unit **108** forms the focusing adjustment information based on a comparison between the locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of the at least two illuminated spots **126A**, **126B** and a reference (X.sub.A0, Y.sub.A0), (X.sub.B0, Y.sub.B0), the reference correspond to the state where the retina **154** is in-focus on the examination detector cell **302**.

(54) In an embodiment, the data processing unit **108** may form the focusing adjustment information based on a comparison between the geometric characteristic of the at least two illuminated spots **126A**, **126B** and the reference characteristic, which corresponds to the state of being in-focus. The state of being in-focus may refer to the state where the retina **154** is in-focus on the examination detector cell **302**. The data processing unit **108** may include, in the focusing adjustment information, data on a property of the deviation from the reference characteristic i.e. a deviation from a proper focus. The property of the deviation may comprise one of the following: the reference characteristic differs in one manner from the geometric characteristic of the at least two illuminated spots **126A**, **126B**, **126C** and the reference characteristic differs in an opposite manner from the geometric characteristic of the at least two illuminated spots **126A**, **126B**. The constellation of the illuminated spots **126A**, **126B**, **126C** may be too large or too small with respect to that of the reference characteristic. In a simpler case, the distance D of the illuminated spots **126A**, **126B** may be too large or too small with respect to that of the reference characteristic. The difference between the geometric characteristic and the reference characteristic may vary according to linear or non-linear function with respect to the variation of the refraction. The function may be monotonic in the used diopter range. Some examples of such functions are linear equation, or polynomial-function. The function may be piecewise linear or piecewise polynomial function, too.

(55) In an embodiment, the data processing unit **108** may form the focusing adjustment information based on a comparison between the at least one distance **D** between the at least two illuminated spots **126A**, **126B** and a reference distance **D0**, which corresponds to the state where the retina **154** is in-focus on the examination detector cell **302**. Then the data processing unit **108** may include, in the focusing adjustment information, data on a direction of the deviation, the direction of the deviation comprising one of the following: the reference distance **D0** is larger than the at least one distance **D** between the at least two illuminated spots **126A**, **126B** and the reference distance **D0** is smaller than the at least one distance **D** between the at least two illuminated spots **126A**, **126B**. Based on this information, the refraction power of the imaging system **300** of the ophthalmic examination apparatus can be increased or decreased for forming an image of the retina **154** that is in-focus on the examination detector cell **302**.

(56) In an embodiment an example of which is illustrated in FIGS. **3** and **5**, the infrared light source system **100** may direct the infrared light to the retina **154** through at least one sub-section **140**, **140A**, **140B** of the pupil **152** of the eye **150**. The receiving arrangement **102** receives the infrared light reflected from the retina **154** through at least two areas **122A**, **122B** of the pupil **152** of the eye **150**. The at least one sub-section **140**, **140A**, **140B** and the at least two areas **122A**, **122B** may be non-overlapping or at least separate and at least almost non-overlapping.

(57) In an embodiment an example of which is illustrated in FIG. **3**, the receiving arrangement **102** may comprise a second beam splitter **202**. The second beam splitter **202** may perform one of the following: reflect the at least a part of the infrared light reflected from the retina **154** to the at least two apertures **120A**, **120B** (see FIGS. **3**, **4**), and allow at least part of the infrared light reflected from the retina **154** to pass through to the at least two apertures **120A**, **120B** (see FIGS. **1A**, **1B**).

(58) In an embodiment an example of which is illustrated in FIG. **3** the infrared light source system **100** may comprise a source aperture **340**, an imaging component **342** and an imaging aperture **344**. The imaging component **343**, which may be a lens, for example, forms an image of the source aperture at an image plane **350**. Then the image forming arrangement **104** with such as at least one common image forming optical component **210** the forms an image of the source aperture **340** at the retina **154**. The source aperture **340** may include the focusing pattern **130**. The source aperture **340** may have a fixed structure that has the image of the focusing pattern **130**. Additionally or alternatively, the source aperture **340** may comprise a display such as a liquid crystal display, for example. The display may be controlled by the data processing unit **108**. Then the focusing pattern **130** may be static or it may vary in size, location on the retina **154**, color and/or shape as a function of time and/or a person.

(59) The at least one common image forming optical component **210** may comprise one of more optically refractive and/or reflective components which may form a real image or a virtual image. The one of more optically refractive and/or reflective components may comprise at least one lens and/or mirror, which has a curved surface.

(60) In an embodiment an example of which is illustrated in FIG. **4**, the infrared light source system **100** may comprise a mirror **204**. The mirror **204** may receive the infrared light and reflect the infrared light toward the pupil **152** of the eye **150**. The mirror **204** may be located at a non-zero distance from the optical axis **OA** of the receiving arrangement **102** for allowing the infrared light reflected from the retina **154** to propagate without hitting the mirror **204** along the receiving arrangement **102**.

(61) In an embodiment, the infrared light source system **100** may direct light to the eye **150** also directly without the mirror **204** or the first beam splitter **200**. In such a case, the infrared light source system **100** may be at a location of the mirror **204**, for example.

(62) In an embodiment an example of which is illustrated in FIG. **4**, the infrared light source system **100** and the receiving arrangement **102** may share the at least one common image forming optical component **210**, which may comprise at least one lens, for example. The common image forming optical component **210** may form an intermediate image **400** of the retina **154** in an image

plane in front of the second beam splitter **202** (see FIGS. 3, 4). The intermediate image **400** may then be imaged on the examination detector cell **302**. FIGS. 1A, 1B illustrate an example where the intermediate image **400** and the image plane are between the second beam splitter **202** and the detecting surface arrangement **106**, and the image of the retina **154** is directed with the second beam splitter **202** to the examination detector cell **302** before the intermediate image **400** in the image space **110** of the receiving arrangement **102**.

(63) In an embodiment examples of which are illustrated in FIGS. 3 and 4, each of the at least two apertures **120A**, **120B** may be linked to an individual image forming component **160'**, **160''**, which participates in the image formation of the focusing pattern **130** on the detecting surface arrangement **106** and may be considered a part of the receiving arrangement **102** and the image forming arrangement **104**. The image forming components **160'**, **160''** may be lenses and the image forming components **160'**, **160''** may locate between the at least apertures **120A**, **120B** and the detecting surface arrangement **106**.

(64) The individual image forming component **160'**, **160''** may comprise one of more optically refractive and/or reflective components which may form a real image or a virtual image. The one of more optically refractive and/or reflective components may comprise at least one lens and/or mirror, which has a curved surface.

(65) In an embodiment, the receiving arrangement **102** may form an intermediate image of the image plane **400** at an image plane **352**. Then the individual image forming components **160'**, **160''** form images of the intermediate image at the image plane **352** on the detecting surface arrangement **106**. Because the retina **154** is imaged at the image planes **400** and **352**, the focusing pattern **130** is also imaged on the detecting surface arrangement **106**. When the retina **154** is in-focus at the image planes **400**, **352**, the focusing pattern **130** is also in-focus at the detecting surface arrangement **106**.

(66) FIG. 5 illustrates the ophthalmic examination apparatus, which comprises the focus measuring arrangement explained using FIGS. 1A to 4 and an illuminating unit **500**, an imaging system **300** and a focusing actuator **304**. In an embodiment, the imaging system **300** may comprise an image forming system **306**, which participates in the image formation of the retina **154** on the examination detector cell **302**. The image system **306** may comprise at least one lens, for example.

(67) The image forming system **306** may comprise one of more optically refractive and/or reflective components which may form a real image or a virtual image. The one of more optically refractive and/or reflective components may comprise at least one lens and/or mirror, which has a curved surface.

(68) The illumination unit **500** may illuminate the retina **154** through a first region **510** at the pupil **152** of the eye **150**, and the examination detector cell **302** may receive the light reflected from the retina **154** through a second region **512** at the pupil **152** of the eye **150**. The first region **510** and the second region may be non-overlapping at least at two of the following: the cornea of the eye **150** and the first surface of the crystalline lens of the eye **150**, and the second surface of the crystalline lens of the eye **150**. In this manner, the Gullstrand's principle may at least almost be fulfilled.

(69) The illuminating unit **500** may comprise an optical radiation source **502**, which may output visible light, an optical beam forming component **504** and a third beam splitter **506**. Instead of the third beam splitter **506**, the illumination unit **500** may comprise a mirror similar to the mirror **204**.

(70) The illuminating unit **500** may illuminate the retina **154** of the eye **150**. The imaging system **300** comprises the examination detector cell **302**, which has the optically determined relation with respect to the detecting surface arrangement **106**. In an embodiment, the optically determined relation may mean that the detecting surface arrangement **106** and the examination detector cell **302** are optical conjugates. In an embodiment, the detecting surface arrangement **106** is configured to detect infrared light and the examination detector cell **302** is configured to detect visible light, and the detecting surface arrangement **106** and the examination detector cell **302** are optical conjugates such that when the retina **154** is in-focus on the detecting surface arrangement **106** in the infrared light of the focusing pattern **130**, the retina **154** is also in-focus on the examination

detector cell **302** in the visible light used in the examination of the eye **150**. In this manner, although a proper focus is found, the retina **154** may be in-focus on one of the detecting surface arrangement **106** and the examination detector cell **302** and simultaneously out-of-focus on another of the detecting surface arrangement **106** and the examination detector cell **302** if measured or considered in one wavelength only. But when measured or considered in two different wavelengths that are expressly used for the detecting surface arrangement **106** and the examination detector cell **302** they both can be simultaneously in-focus. The reason for this is that a refraction index of a lens is different for different wavelengths, for example.

(71) The focusing actuator **304** receives the focusing adjustment information, and adjusts the imaging system **300** optically to form a focused image of the retina **154** onto the examination detector cell **302** based on the focusing adjustment information. The focusing actuator **304** may also adjust the at least one common image forming optical component **210**. In an embodiment, the focusing actuator **304** may change a distance between at least two lenses of the imaging system **300** and/or the at least one common image forming optical component **210** for causing the image of the retina **154** to be in-focus on the examination detector cell **302**. In an embodiment, the focusing actuator **304** may change a refraction index of at least one lens the imaging system **300** and/or the at least one common image forming optical component **210** for causing the image of the retina **154** to be in-focus on the examination detector cell **302**. The at least one common image forming optical component **210** may be considered a part of the imaging system **300** but it may also be considered a part of the illumination system providing the visible light to the eye **150**, the receiving arrangement **102** and/or the image forming arrangement **104**.

(72) The focusing actuator **304** may adjust the focal length of the imaging optics, and/or a distance between the imaging optics and the imaging plane. The adjustment may be performed alternatively, successively or simultaneously. The focal length adjustment may be performed by adjusting distance between at least two optical elements, such as lenses or mirrors, or by at least one electrically tunable lens, such as liquid lens, the refractive index or the curvature of which is changed, for example.

(73) In an embodiment, the focusing actuator **304** may also adjust the individual image forming components **160'**, **160''** for forming the image of the focusing pattern **130** that is in-focus on the detecting surface arrangement **106**.

(74) In an embodiment examples of which are illustrated in FIGS. **3** and **4**, the second beam splitter **202** may perform one of the following: reflect the at least a part of the illumination reflected from the retina **154** to the examination detector cell **302**, and allow at least part of the infrared light reflected from the retina **154** to pass through to the examination detector cell **302**.

(75) In an embodiment, the examination detector cell **302** and the detecting surface arrangement **106** are included in the same detector. According to this embodiment, there is only one detector which may both receive the infrared light of the focusing pattern **130** passed through the at least two apertures **120A**, **120B** and the visible light reflected from the retina **154**, the visible light being used for the ophthalmic examination.

(76) In an embodiment examples of which are illustrated in FIGS. **3** and **4**, the focus measuring arrangement may comprise a common aperture stop **308**, which is common to the examination detector cell **302** and the detecting surface arrangement **106**. The at least two apertures **120A**, **120B** may receive the infrared light of the focusing pattern **130** from the retina **154** through separate sections **310A**, **310B** of the common aperture stop **308**. The separate sections **310A**, **310B**, in turn, may be non-overlapping. The image forming system **104** may then cause an optical correspondence between the at least two apertures **120A**, **120B** and the separate sections **310A**, **310B** of the common aperture stop **308**. That is, the image forming system **104** forms, in practice and/or in theory, images of the at least two apertures **120A**, **120B** on the common aperture stop **308** and the images of the at least two apertures **120A**, **120B** are the separate sections **310A**, **310B**. The at least two apertures **120A**, **120B** and thus the separate sections **310A**, **310B** may correspond optically the

areas **122A**, **122B** on the pupil **152** of the eye **150**.

(77) FIG. **6A** illustrates an example of paths of the infrared light of the focusing pattern **130**, which goes through the separate sections **310A**, **310B** at the aperture stop **308**. The separate sections **310A**, **310B** are images **122A**, **122B** of at least two apertures **120A**, **120B** at the pupil plane **152'** and/or the common aperture stop **308**. FIG. **6A** illustrates the paths seen in a direction perpendicular to a normal of the pupil plane **152'**.

(78) FIG. **6B** illustrates an example of paths of the infrared light of the focusing pattern **130**, which goes through separate sections **310A**, **310B**. FIG. **6B** illustrates the paths seen in a direction of the normal of the pupil plane **152'**.

(79) FIG. **7A** to **7D** illustrate various shapes of the focusing patterns **310** without limiting to these. The focusing patterns **310** are considered to be on the retina **154** in FIGS. **7A** to **7D**. Although the focusing patterns **310** are illustrated as sharp images, the focusing patterns **310** may alternatively be more or less blurred.

(80) FIG. **8** illustrates an example of how rays of the infrared light propagate from the intermediate image **402** to the detecting surface arrangement **106**. The detecting surface arrangement **106** may be a line sensor or a matrix sensor. Three rays are depicted in each three different focus settings, which shows changes in a position of the intermediate image **402** along the optical axis **OA**. In the first situation, the intermediate image **402** is formed too far from the detecting surface arrangement **106** and it causes a distance **D1** between the two illuminated spots **126A**, **126B**. In this example of FIG. **8**, the illuminated spots **126A**, **126B** are blurred because they are out-of-focus. In the second situation, the intermediate image **402** is formed at a desired image plane **400** for the detecting surface arrangement **106** and it causes a distance **D0** between the two illuminated spots **126A**, **126B**. In this case the illuminated spots **126A**, **126B** are focused at the detecting surface arrangement **106**. In the third situation, the intermediate image **402** is formed close to the detecting surface arrangement **106** and it causes a distance **D2** between the two illuminated spots **126A**, **126B**. In this example of FIG. **8**, the illuminated spots **126A**, **126B** are blurred because they are out-of-focus in cases of FAR and NEAR.

(81) When the individual images **124A**, **124B** of the focusing pattern **130** are formed at the detecting surface arrangement **106** in an image space **110** of the receiving arrangement **102**, the illuminated spots **126A**, **126B** are in-focus (see spot at the distance **D0** from each other in FIG. **8**).

(82) FIG. **9A** to **9D** illustrate how different shapes focusing patterns **130** can be utilized. FIGS. **9A** to **9D** show the focusing patterns **130** at the retina **154** on left, and the separate areas **122A**, **122B**, **122C** at the pupil **152** on right. The pictures on right may also be considered or they have a certain correspondence with images of the apertures **120A**, **120B**, **120C** at the pupil **152**, at the pupil plane **152'**, the common aperture stop **308** and/or at the detecting surface arrangement **106**. At the pupil **152** and the pupil plane **152'** the images are the areas **122A**, **122B**, **122C**. At the common aperture **308** the images are the separate sections **310A**, **310B**, **310C**. At the detecting surface arrangement **106** the images are the illuminated spots **126A**, **126B**, **126C**.

(83) FIGS. **9E** to **9G** illustrate examples of illuminated spots **126A** to **126C** of various focusing patterns at the detecting surface arrangement **106** and the detecting surfaces **128A** to **128C**. FIGS. **9A** and **9B** correspond to FIG. **9E**. In a similar manner FIGS. **9C** and **9D** correspond to the FIGS. **9F** and **9G**, respectively.

(84) When an elongated focusing pattern **130**, the longitudinal axis of which is vertical, like that in FIG. **9A**, is used, the locations are sensitive to the horizontal movement. The beams of infrared light may have a large diameter and the beams may almost completely fill the pupil **152**, which is illustrated in FIG. **9B**. FIG. **9C** illustrates a pair of the focusing patterns **130**. One of the focusing patterns may be elongated in a vertical direction, and the infrared light of it may pass the pupil **152** through areas that are side by side at the pupil **152** of the eye **150**. Another of the focusing patterns may be elongated in a horizontal direction, and the infrared light of it may pass the pupil **152** through areas where one is above another at the pupil **152** of the eye **150** in the vertical direction.

FIG. 9D illustrates an example of the focusing pattern **130** that is a circle. In this example there are three apertures **120A**, **120B** and **120C** images of which are distributed circularly around the pupil **152** of the eye **150** in a non-overlapping manner or at least almost non-overlapping manner. When the detecting surface arrangement **106** has a two-dimensional matrix sensor, the variation of locations of the illuminated spots **126A**, **126B**, **126C** can be monitored in three different directions. It means that the eye astigmatism can be fully measured, i.e. both in its magnitude and axis orientation.

(85) The image forming arrangement **104** images the retina **154** to the detecting surface arrangement **106** such that centroids or the like of images of the separate areas **122A**, **122B**, which are caused by the at least two apertures **120A**, **120B**, at the pupil plane **152'** and on the detecting surface arrangement **106** deviate from each other. The deviation can be denoted by a vector $v_{sub.i}$ on the pupil plane **152'**. Each beam of the infrared light that travels through one of the separate areas **122A**, **122B** has typically at least one such a paired beam of the infrared light in another of the separate areas **122A**, **122B**, that the distance D between the centroids of the pair $|v_{sub.i}|$ is larger than approximately a quarter of the smallest diameter of the pupil **152**, or for example approximately a half of or larger than the smallest diameter of the pupil **152**. However, the distance D may be (but not necessarily) smaller than the largest diameter of the pupil **152**. The centroids of the images of the separate areas **122A**, **122B** may locate approximately inside the projection of the pupil **152** on the pupil plane **152'**.

(86) The focal shift calculated for a pair of illuminated spots **126A**, **126B**, for example, may reveal a required focusing need in a direction of the deviation vector $v_{sub.i}$ corresponding to a pair of illuminated spots **126A**, **126B** (see FIGS. 8, 9A to 9D). The vector $v_{sub.i}$ has a magnitude and direction. The direction of the deviation vector $v_{sub.i}$ may point to left or right, for example. The direction of the deviation vector $v_{sub.i}$ may point to up or down, for example. More generally, the direction of the deviation vector $v_{sub.i}$ may point to one direction or to an opposite direction, for example. The ophthalmic instrument may contain at least three pairs of analysis beams formed by at least three apertures similar to the apertures **120A**, **120B**, whose deviation vectors $v_{sub.i}$ are substantially non-parallel, such that astigmatism of the eye **150** may be caused to result in different focus shifts between the illuminated spots of the pairs of beams. That information can be used for focusing optimally even the eye **150** having astigmatism.

(87) The focus measuring arrangement may be arranged to form pairs of analysis beams with the apertures **120A**, **120B** and the deviation vectors $v_{sub.i}$ based on the illuminated spots **126A**, **126B**, **126C** in any direction where focusing information is needed for the optimal focusing of the ophthalmic examination apparatus.

(88) As the focusing measurement gives not only the direction to which the focus needs to be adjusted, but also the amount of the needed focus adjustment, the described focusing method and adjustment may be performed fast. This may be a useful feature, since the eye **150** under examination tends to get tired quickly and starts drying and moving. One focusing round may be enough for an accurate focus.

(89) The focusing measurement and adjustment may be performed accurately, robustly and in a reliable manner by the following two aspects:

(90) The focus pattern **130** may be arranged to locate in one or more areas on the retina **154**, where the most accurate focusing is required for optimal use of the ophthalmic instrument. In other words, the focusing pattern **130** may comprise spots (or for example line segments) in several different positions on the retina **150**, and the focus information can be calculated from those different positions at the same time.

(91) The infrared light may be arranged to travel through the front parts of the eye **150** (from crystalline lens to cornea) within such a path, that it optimally takes account the deformations in the person's eye **150** for the ophthalmic instrument operation. For example, in the fundus cameras the infrared light may span the imaging infrared light travelling toward the fundus camera such that the

infrared light travelling into the eye **150** experiences the same out-of-focus state and aberrations as the imaging infrared light travelling out of the eye **150**. In this manner, the focusing pattern **130** of the infrared light may help in finding the focus with a precise manner.

(92) The robust and accurate operation may be achieved so that the optical beams of the focusing pattern **130** to the detecting surface arrangement **106** are part, i.e. subset, of the beam from the retina **154** to the examination detector cell **302**, all the way through the path from the retina **154** to the second beam splitter **202**. In that manner, the beams which measure the focus, and the beam which is used to examine the retina **154**, experience maximally the same eye accommodation, the same eye aberrations, and the same aberrations and tolerances through the instrument optics. Hence, it may be assumed that the measured focus corresponds the most accurately to the best focus for the examination detector cell **302**.

(93) The electric signal levels of the detecting surface arrangement **106** may be amplified with the hardware design if necessary for improving the focus measurement.

(94) Here is the focus measuring arrangement for ophthalmic examination in a nutshell. As the optical beams of the focusing pattern **130** are detected based on the non-telecentric manner, the locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of the spots **126A**, **126B** (possibly out-of-focus) of the focusing pattern **130** on the detecting surface arrangement **106** depends on the focus of the focus measuring arrangement which has a deterministic relation to the focus of the ophthalmic examination apparatus.

(95) A direction and a magnitude related to the locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B) and/or geometric difference of the (possibly out-of-focus) images of the focusing pattern **130** from a reference (X.sub.A0, Y.sub.A0), (X.sub.B0, Y.sub.B0) can be computed, and determine a geometric difference between the images and the reference. The difference based on the direction and the magnitude, for example, may be understood to be a phase difference. The difference may be determined pairwise of the illuminated spots and one or more distance-value may be formed.

(96) Due to the non-telecentricity at the receiving arrangement **102**, a value of the location and/or the geometric difference may be a monotonic function of the required focal shift and it may be used to focus the retina **154** at the detecting surface arrangement **106** and/or the examination detecting cell **302** based on the deviation vector $v_{sub.i}$ between the at least one pair of the illuminated spots **126A**, **126B**, **126C**. The function may be determined by a calibration of the ophthalmic instrument where the reference is determined when the ophthalmic examination apparatus is in-focus.

(97) The presented solution may be used in many kinds of ophthalmic instruments, such as ophthalmic treatment instruments, for example. This focusing arrangement and method may be suitable to be used in any instrument or device which needs to be in-focus with the eye **150**.

(98) An operator may be a person, but the ophthalmic instrument may also be an automatic or autonomous device where the operator is replaced by automation. The operator may also not be present next to the ophthalmic instrument, but may be in a different location and operate the instrument by a remote access.

(99) The ophthalmic instrument can be either a handheld device or a desktop device. The design may be scalable in terms of size. For example, handheld ophthalmic instruments tend to be as small in a physical size as possible and a miniaturized design is more feasible there.

(100) Although a single focusing point is described above, there may be multiple focusing beams, which are targeted to multiple focusing points at the retina **154**, so that the focus may be measured from several positions at the retina **154**. That may help determining a best overall focus over the whole field-of-view of the ophthalmic instrument.

(101) The focusing arrangement and method described here can be combined with contrast detection based autofocus methods, for example, such that both of the methods can be used successively, or such that a method, which fits better to the task under action, may be selected.

(102) The ophthalmic instrument may use a wavelength band for the eye examination and a different wavelength band for finding focus for it. For example, the ophthalmic examination

apparatus such as a fundus camera may capture images of the retina **154** by using a visible wavelength band and the focus measuring arrangement may use near-infrared wavelengths for the focusing purposes. The required focus difference between the visible and near infrared wavelengths may be known based on calibration, simulation and/or theoretical calculations such that the difference can be taken account when algorithm calculates the required focusing adjustment.

(103) FIG. **10** an example of the data processing unit **108**, which may comprise one or more processors **1000** and one or more memories **1002** including computer program code. The one or more memories **1002** and the computer program code, with the one or more processors **1000** may cause the focus measuring arrangement at least to perform the method illustrated in FIG. **11**. In a corresponding manner, the data processing unit **108** be a part of the ophthalmic examination apparatus where the one or more memories **1002** and the computer program code, with the one or more processors **1000** may cause the ophthalmic examination apparatus at least to perform the method illustrated in FIG. **12**.

(104) FIG. **11** illustrates the focus measuring method of ophthalmic examination. In step **1100**, infrared light is directed to a pupil **152** of an eye **150** of a person with an infrared light source system **100**.

(105) In step **1102**, a focusing pattern **130** of the infrared light is formed on a retina **154** of the eye **150**.

(106) In step **1104**, the infrared light of the focusing pattern is received from the retina **154** through separate areas **122A**, **122B** of the pupil **152** with at least two apertures **120A**, **120B** of a receiving arrangement **102**.

(107) In step **1106**, an individual image **124A**, **124B** of the focusing pattern **130** is formed with an image forming arrangement **104**, for each of at least two apertures **120A**, **120B**, which the infrared light passed through, in an image space **110** of the receiving arrangement **102**, while the image forming arrangement **104** causes an optical correspondence between the at least two apertures **120A**, **120B** and the separate areas **122A**, **122B** of the pupil **152**.

(108) In step **1108**, receiving the infrared light passed through the at least two apertures **120A**, **120B** with a detecting surface arrangement **106** located at a non-zero distance behind the at least two apertures **120A**, **120B** and the image forming arrangement **104** in a direction of propagation of the infrared light reflected from the retina **104**.

(109) In step **1110**, focusing adjustment information is formed with a data processing unit **108** based on locations (X.sub.A, Y.sub.A), (X.sub.B, Y.sub.B), (X.sub.C, Y.sub.C) of illuminated spots **126A**, **126B**, **126C** of the infrared light of the focusing pattern **130** on the detecting surface arrangement **106**.

(110) In step **1112**, the data processing unit **108** outputs the focusing adjustment information for further processing.

(111) FIG. **12** illustrates the ophthalmic examination method. In step **1200**, the focus measuring method of FIG. **11** is performed.

(112) In step **1202**, the retina **154** of the eye **150** is illuminated with an illuminating unit **500**.

(113) In step **1204**, light of illuminating unit **500** reflected from the retina **154** is collected with imaging system **300**, which comprises an examination detector cell **302**, which has an optically determined relation with respect to the detecting surface arrangement **106**.

(114) In step **1206**, the focusing adjustment information is received by a focusing actuator **304**.

(115) In step **1208**, the imaging system **300** is adjusted optically to form a focused image of the retina **154** onto the examination detector cell **302** in response to the focusing adjustment information.

(116) The data processing parts of the methods shown in FIGS. **11** and **12** may be implemented as a logic circuit solution or computer program. The computer program may be placed on a computer program distribution means for the distribution thereof. The computer program distribution means is readable by a data processing device, and it encodes the computer program commands, carries

out the measurements and optionally controls the processes on the basis of the measurements.

(117) The computer program may be distributed using a distribution medium which may be any medium readable by the controller. The medium may be a program storage medium, a memory, a software distribution package, or a compressed software package. In some cases, the distribution may be performed using at least one of the following: a near field communication signal, a short distance signal, and a telecommunications signal.

(118) What is presented in this document teaches how to focus the ophthalmic instrument to the eye **150** effectively, and it may result in one or more of the following potential advantages: faster focusing more accurate focusing more accurate focusing in presence of astigmatism in the eye more reliable focusing more user friendly focusing.

(119) It will be obvious to a person skilled in the art that, as technology advances, the inventive concept can be implemented in various ways. The invention and its embodiments are not limited to the example embodiments described above but may vary within the scope of the claims.

Claims

1. A focus measuring arrangement for ophthalmic examination, which comprises: an infrared light source system, which is configured to direct infrared light to a pupil of an eye of a person and form a focusing pattern of the infrared light on a retina of the eye, wherein the focus measuring arrangement further comprises a receiving arrangement, which comprises at least two apertures, which are configured to receive the infrared light of the focusing pattern from the retina through separate areas of the pupil, an image forming arrangement, which is configured to cause the infrared light passed through the at least two apertures to form an individual image of the focusing pattern for each of the at least two apertures in an image space of the receiving arrangement, the image forming arrangement being configured to cause an optical correspondence between the at least two apertures and the separate areas of the pupil, and a detecting surface arrangement located at a non-zero distance behind the at least two apertures and the image forming arrangement in a direction of propagation of the infrared light reflected from the retina for receiving the infrared light passed through the at least two apertures; and a data processing unit, which is configured to form focusing adjustment information based on locations of illuminated spots of the infrared light of the focusing pattern on the detecting surface arrangement, and output the focusing adjustment information for further processing.

2. The focus measuring arrangement of claim 1, wherein the data processing unit is configured to form the focusing adjustment information based on a geometric characteristic of the at least two illuminated spots on the detecting surface arrangement, the characteristic being a function of deviation from a state of being in-focus of the focus measuring arrangement.

3. The focus measuring arrangement of claim 2, wherein the geometric characteristic is at least one distance between the at least two illuminated spots on the detecting surface arrangement, the distance varying as a function of deviation from a state of being in-focus of the focus measuring arrangement.

4. The focus measuring arrangement of claim 2, wherein the data processing unit is configured to form the focusing adjustment information based on a comparison between the geometric characteristic of the at least two illuminated spots and a reference characteristic, which corresponds to the state of being in-focus, and the data processing unit is configured include in the focusing adjustment information data on a property of the deviation, the property of the deviation comprising one of the following: the reference characteristic differs in one manner from the geometric characteristic of the at least two illuminated spots and the reference characteristic differs in an opposite manner from the geometric characteristic of the at least two illuminated spots.

5. The focus measuring arrangement of claim 3, wherein the data processing unit is configured to form the focusing adjustment information based on a comparison between the at least one distance

between the at least two illuminated spots and a reference distance, which corresponds to the state of being in-focus, and the data processing unit is configured include in the focusing adjustment information data on a direction of the deviation, the direction of the deviation comprising one of the following: the reference distance is larger than the at least one distance between the at least two illuminated spots is smaller than the at least one distance between the at least two illuminated spots.

6. The focus measuring arrangement of claim 1, wherein the infrared light source system is configured to direct the infrared light to the retina through at least one first sub-section of the pupil of the eye; the receiving arrangement is configured to receive the infrared light reflected from the retina through at least one second sub-section of the pupil of the eye; the at least one first sub-section and the at least one second sub-section being non-overlapping.

7. The focus measuring arrangement of claim 1, wherein the receiving arrangement comprises a second beam splitter, which is configured to perform one of the following: reflect the at least a part of the infrared light reflected from the retina to the at least two apertures, and allow at least part of the infrared light reflected from the retina to pass through to the at least two apertures.

8. The focus measuring arrangement of claim 1, wherein each of the at least two apertures is linked to an individual image forming component, which is configured to participate in an image formation of the focusing pattern on the detecting surface arrangement.

9. An ophthalmic examination apparatus, wherein ophthalmic examination apparatus comprises: the focus measuring arrangement of claim 1; and an illuminating unit configured to illuminate the retina of the eye; an imaging system, which comprises an examination detector cell, which has an optically determined relation with respect to the detecting surface arrangement; a focusing actuator, which is configured to receive the focusing adjustment information, and adjust the imaging system optically to form a focused image of the retina onto the examination detector cell in response to the focusing adjustment information.

10. The apparatus of claim 9, wherein the second beam splitter is configured to perform one of the following: reflect the at least a part of the illumination reflected from the retina to the examination detector cell, and allow at least part of the infrared light reflected from the retina to pass through to the examination detector cell.

11. The apparatus of claim 9, wherein imaging system comprises the image forming system.

12. The apparatus of claim 9, wherein imaging system, which comprises the examination detector cell, which is an optical conjugate to the detecting surface arrangement.

13. The apparatus of claim 9, wherein the focus measuring arrangement comprises a common aperture stop, which is common to the examination detector cell and the detecting surface arrangement, and the at least two apertures are configured to receive the infrared light of the focusing pattern from the retina through separate sections of the common aperture stop, the separate sections being non-overlapping, and the image forming system being configured to cause an optical correspondence between the at least two apertures and the separate sections of the common aperture stop.

14. A focus measuring method of ophthalmic examination, the method comprising directing, with an infrared light source system, infrared light to a pupil of an eye of a person, the method comprising forming a focusing pattern of the infrared light on a retina of the eye; receiving, with at least two apertures of a receiving arrangement, the infrared light of the focusing pattern from the retina through separate areas of the pupil; forming, with an image forming arrangement, an individual image of the focusing pattern for each of at least two apertures, which the infrared light passed through, in an image space of the receiving arrangement, while the image forming arrangement causes an optical correspondence between the at least two apertures and the separate areas of the pupil; receiving, with a detecting surface arrangement located at a non-zero distance behind the at least two apertures and the image forming arrangement in a direction of propagation of the infrared light reflected from the retina, the infrared light passed through the at least two apertures; forming, with a data processing unit, focusing adjustment information based on locations

of illuminated spots of the infrared light of the focusing pattern on the detecting surface arrangement; and outputting, from a data processing unit, the focusing adjustment information for further processing.

15. An ophthalmic examination method, the method further comprising performing the focus measuring method of claim 14; and illuminating, with an illuminating unit, the retina of the eye; collecting light of illuminating unit reflected from the retina with imaging system, which comprises an examination detector cell, which has an optically determined relation with respect to the detecting surface arrangement; receiving the focusing adjustment information by a focusing actuator; and adjusting the imaging system optically to form a focused image of the retina onto the examination detector cell in response to the focusing adjustment information.
